City Sense

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Abstract

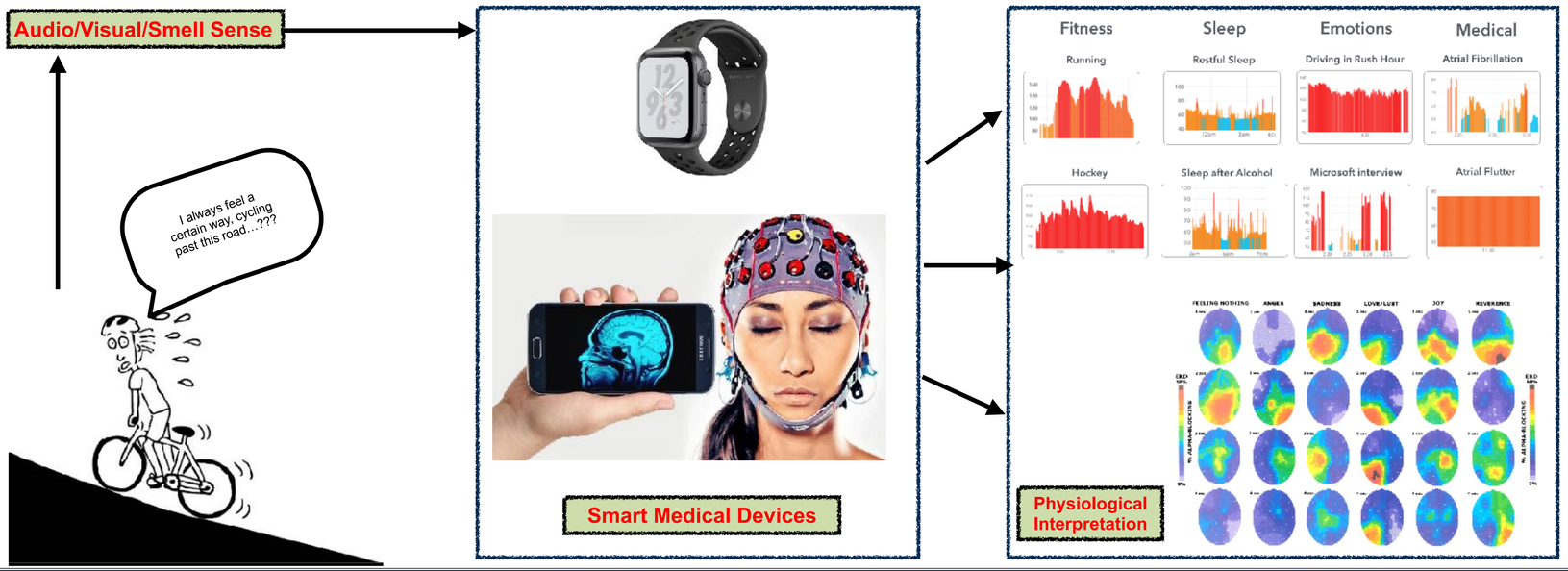
With the medical wearable device market projected to achieve 1.4 billion dollars capitalization by 2020. One would expect that there would be an increase in the number of consumers of such products. Crowd sourcing has proved to be an efficient means of harnessing knowledge cheaply but accurately. The urban land scape is one that is complex in its morphology and evolution. To understand and cater for the complexities that come with our urban areas provide. We propose a framework that leverages personal sensing for urban sensing. If we consider the literature from the fields of neuro-architecture and cognitive geography. We see that the geometry and structural peculiarities of areas actually affect our moods and psychology. Based on this finding, propose to use readings from smart medical devices for the purpose of enriching the geographical information of areas. The benefits of this proposal will include but not limited to catering for . A personalised map for each individual based on their condition. What may be right for you may not be right for the other person. The model will take into account the peculiarities and inform you of the relevant inform you of the

Problem

Our cities are represented using static maps that provide navigation using static algorithms based on shortest path. Whereas a semantic ally enriched map would provide an even better experience for city dwellers. Over 67% of the world's population are projected foo reside in urban centres by 2020. This calls for innovative approaches to accommodate this growth. Back to what we were saying on the complexity of urban dwellings. Factors such as mental health gnu

"we shape our cities and afterwards our cities shape us - anonymous"

Motivation



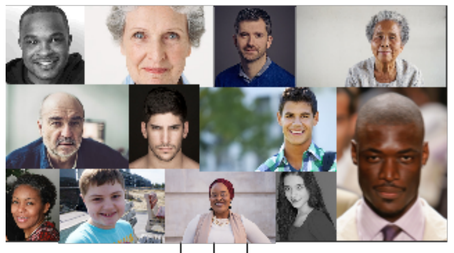
Approach

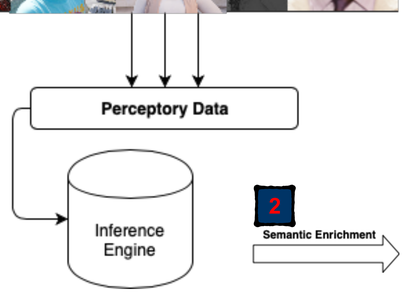
Perceptory data is collected from residents of the city with respect to space. Their physiological response to environmental stimuli is used to develop an affect inference engine. This engine is able to map affect to space and maps this spaces to individual profiles based on health, mental or psychological similarities.

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The semantic enrichment of the city is done via the affective inference. Not only are we able to classify spaces into into affective categorisations. We are also able to personalise the affective experience of the urban dweller based on a profile that has been developed over time. We can offer them the best affective experience they could possibly get.

Objectives & Applications

I. Develop a methodology that accurately constructs and semantifies geographical entities from Geo-spatial data.
II. Define comprehensive taxonomies of geo-spatial entities that reflects the ontology.
III. Represent these taxonomies with efficient data structures. This should be in a form where statistical and computational operations can be performed.
IV. Develop discriminative and optimization models of geographical entities that facilitate accurate semantification.
V. Propose and evaluate an adequate performance evaluation framework.

We can tell residents of places or venues that can lighten their moods based on the changes in recorded users.

We can advise people with a particular health condition say chronic anxiety against visiting certain areas that are notorious for spiking anxiety in indiviudals. E.g autistic kids, older citizens

The truth is that the layout of our buildings and streets has some effect on our mental state. Rebuilding/ reconstruction may be infeasible but we may be able to work around this by enriching our understanding of our living spaces using our physiological responses to urban stimulus.

References

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